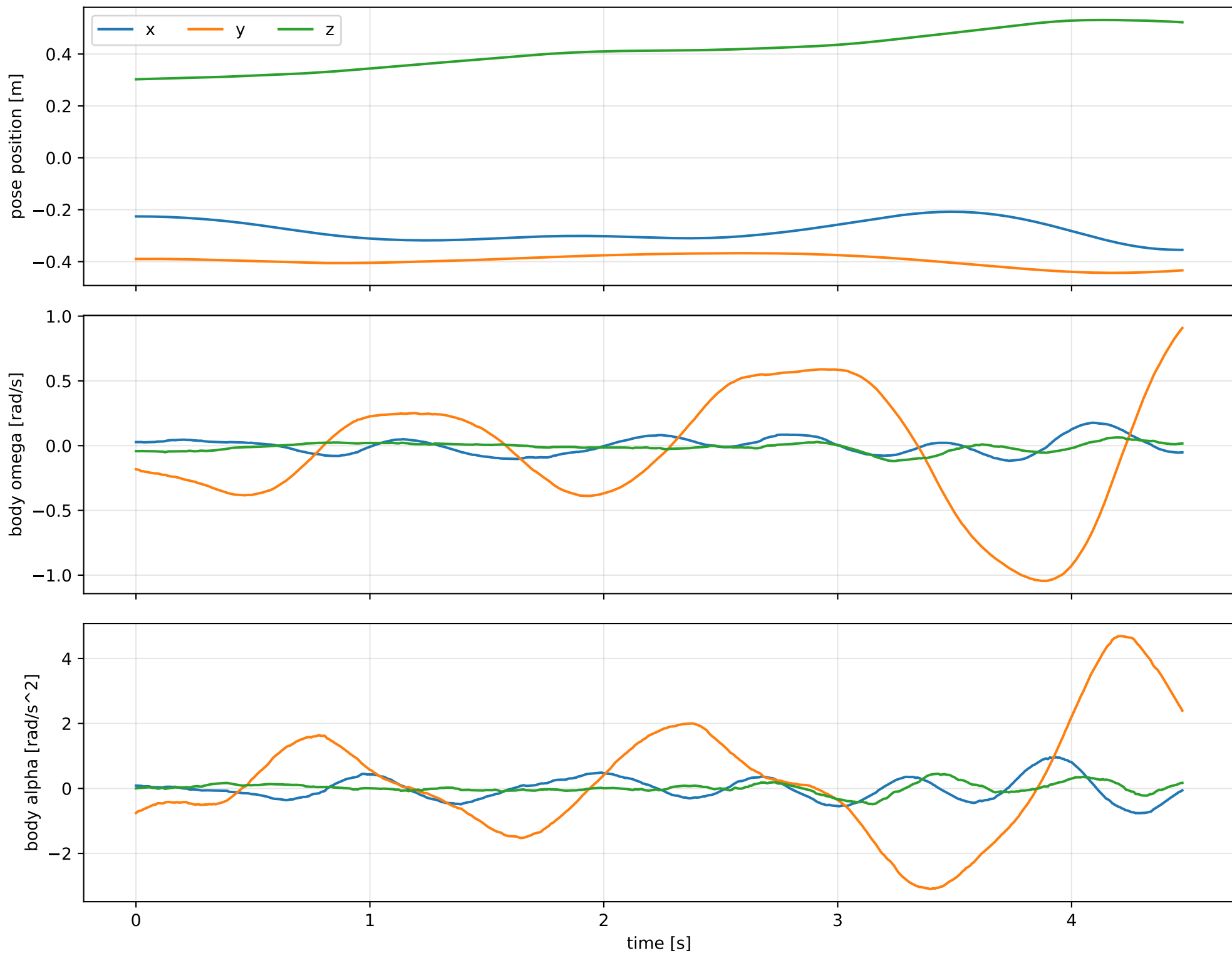
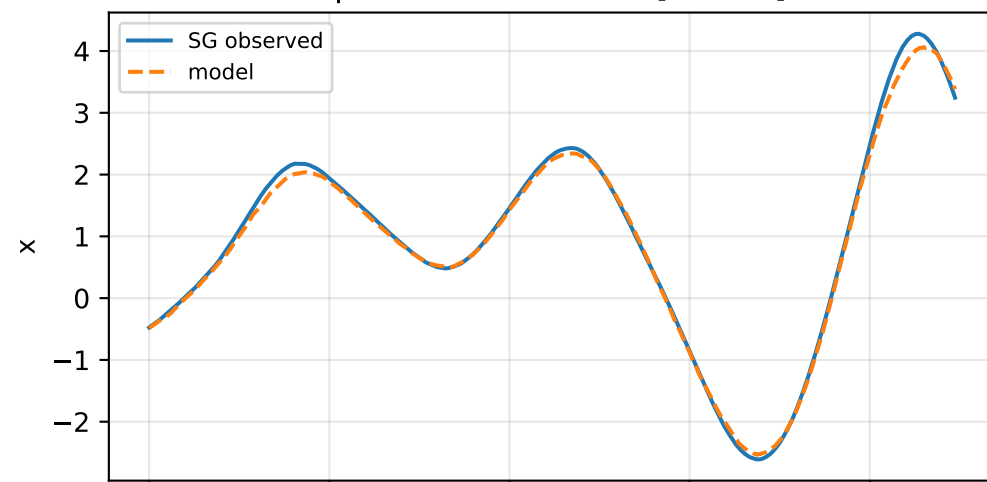


kkt_disabled: SG trajectory and derived motion

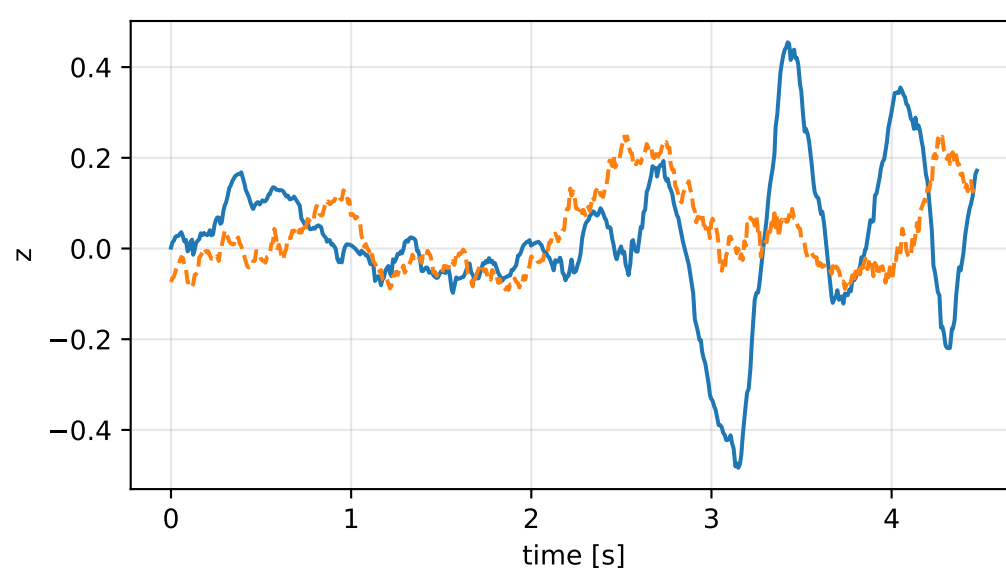
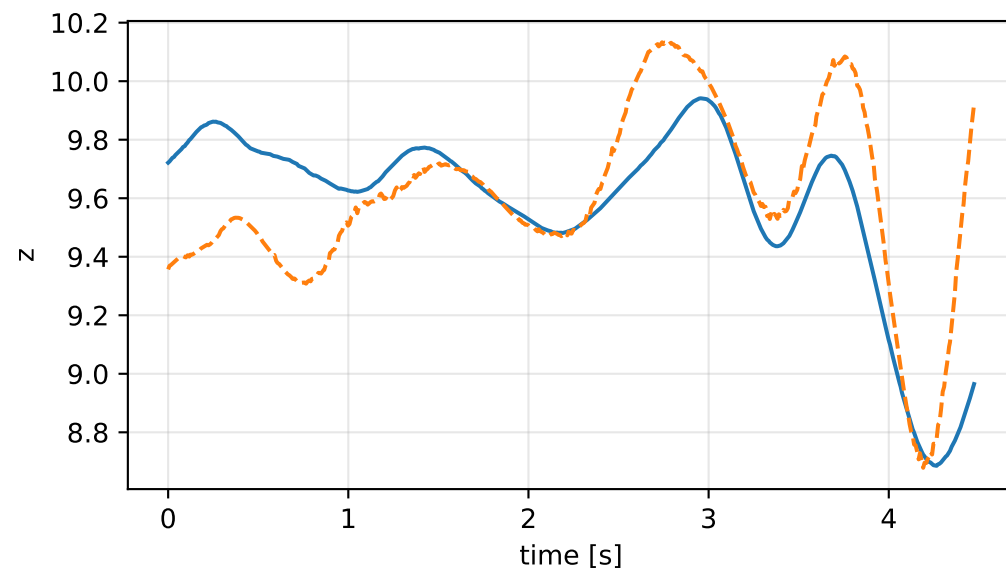
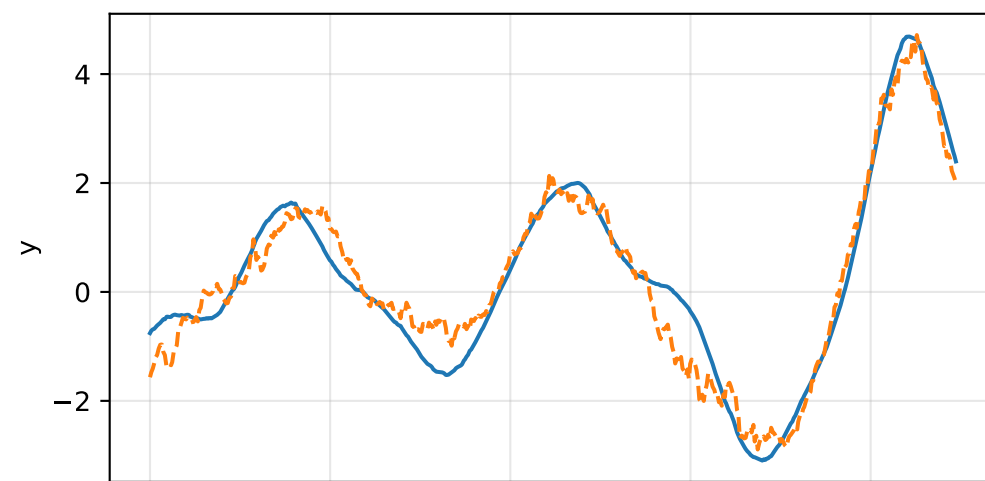
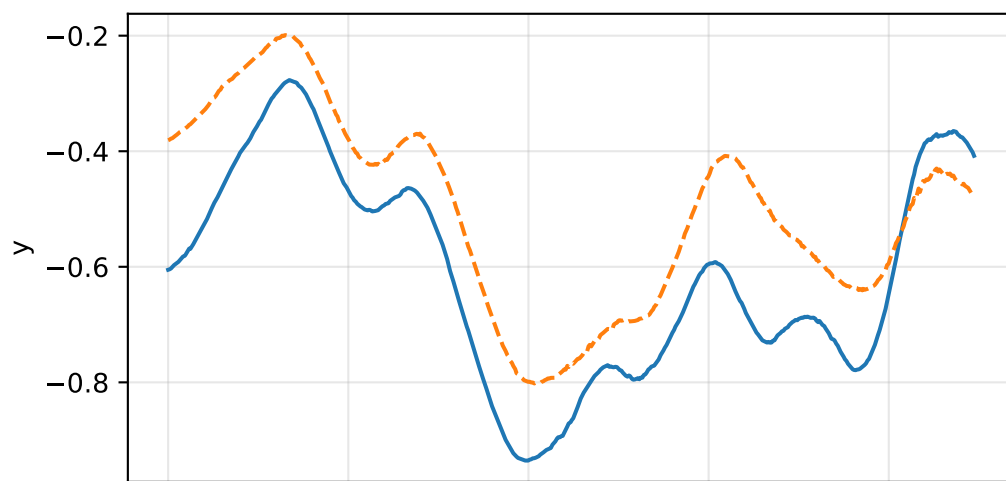
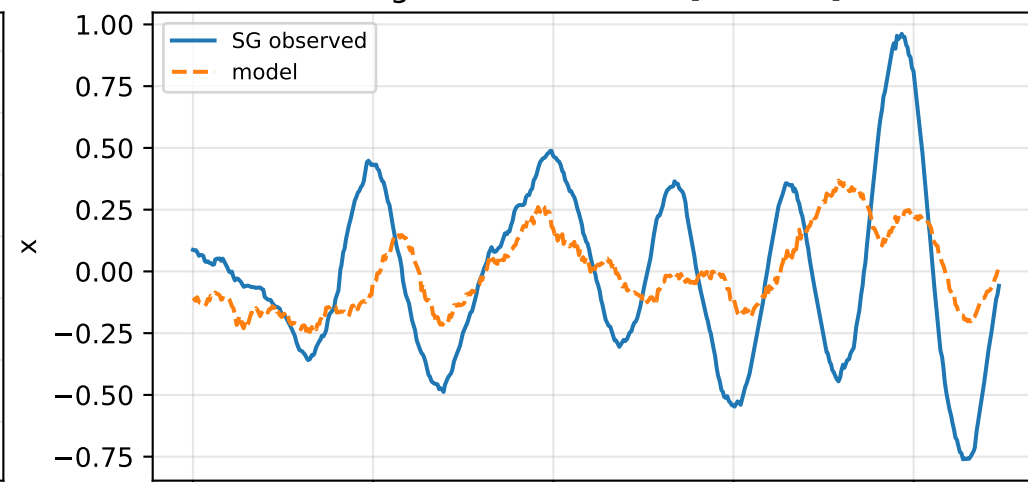


kkt_disabled: acceleration residual objective

specific acceleration [m/s^2]

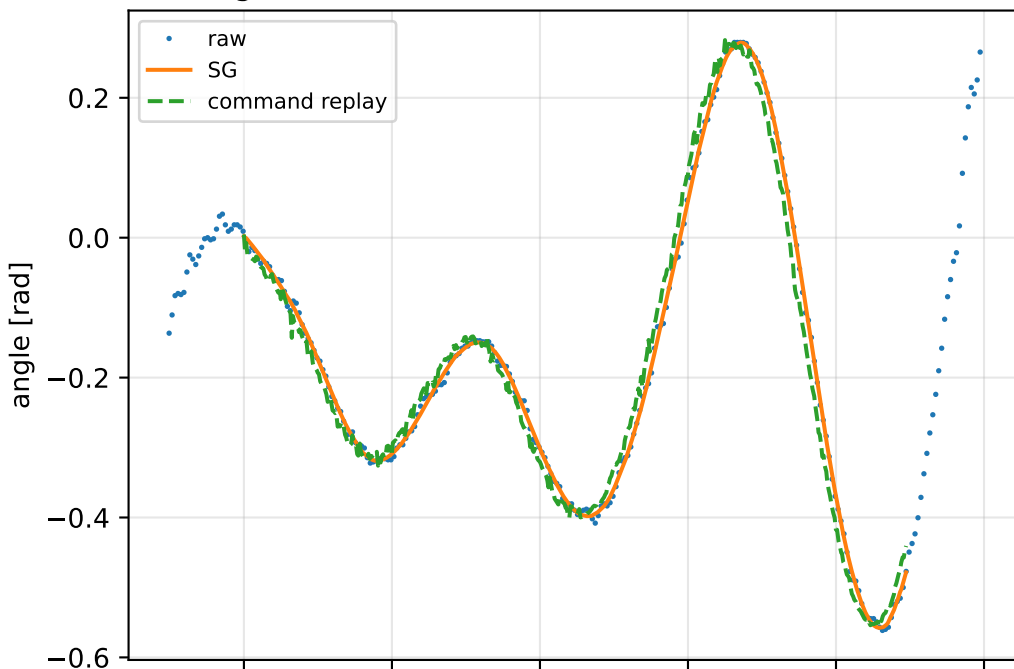


angular acceleration [rad/s^2]

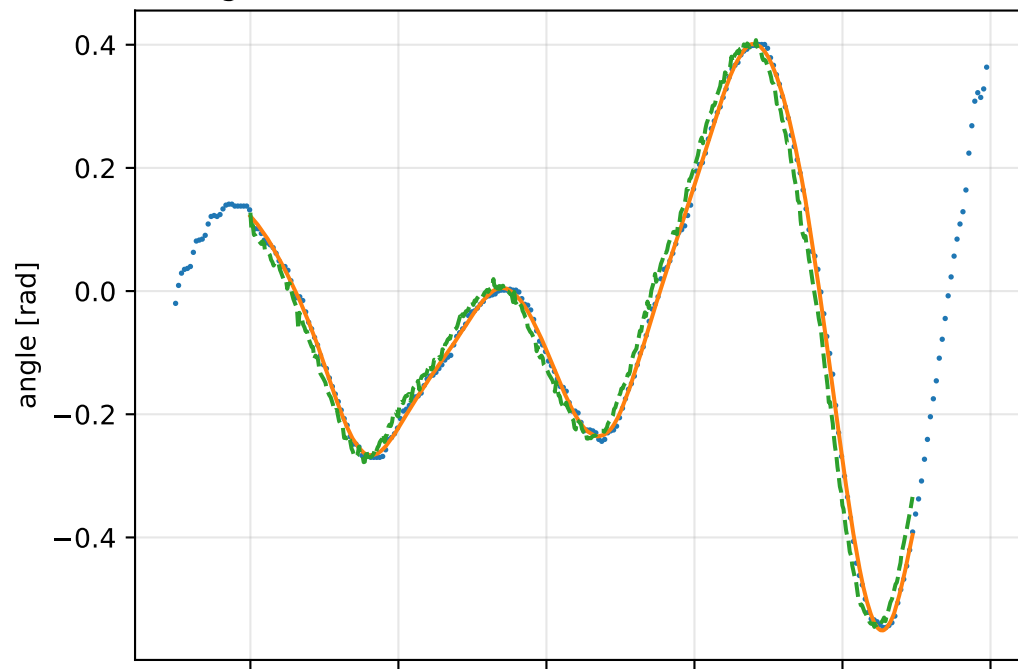


kkt_disabled: gimbal measurement audit (objective=measured_sg)

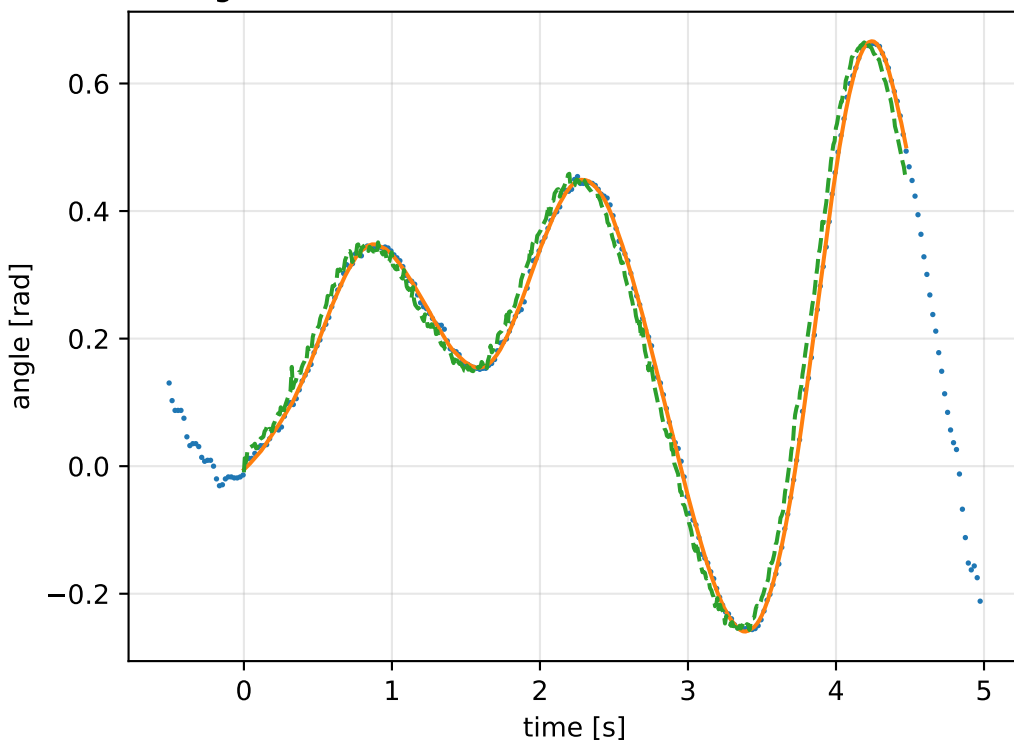
gimbal 1: RMSE=0.027, max=0.07529 rad



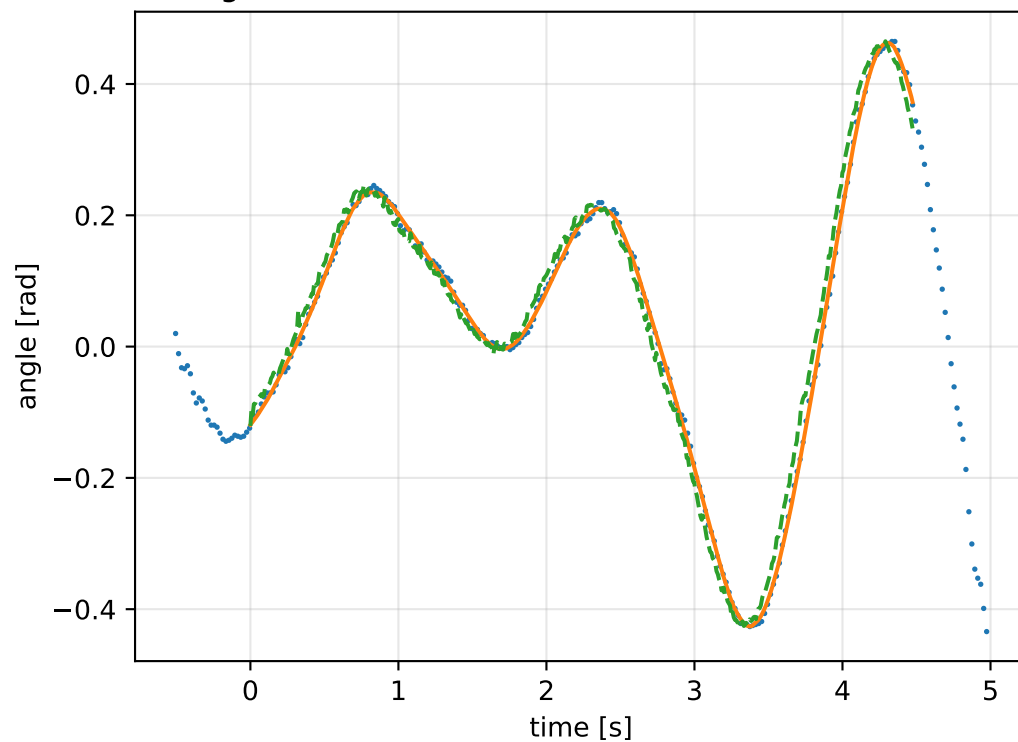
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

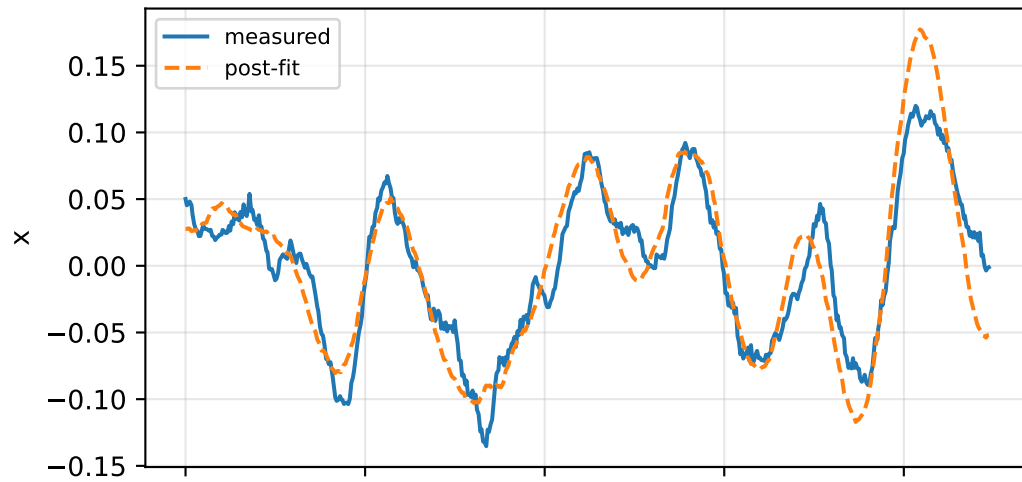


gimbal 4: RMSE=0.02598, max=0.07261 rad

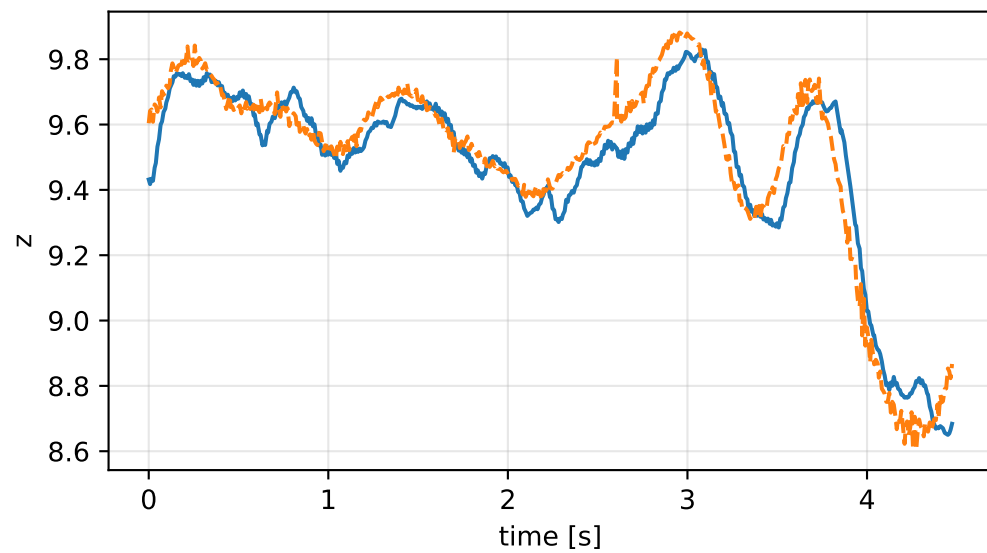
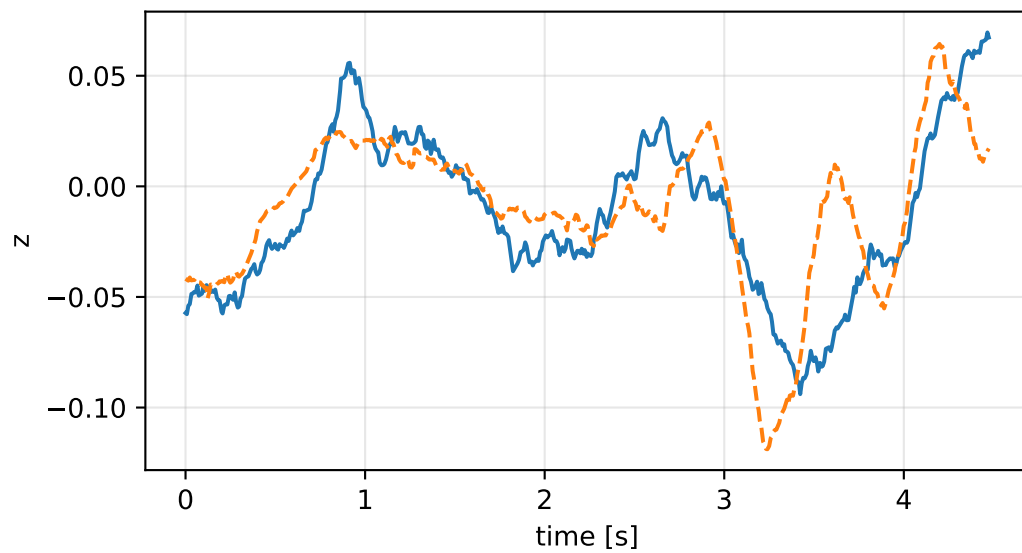
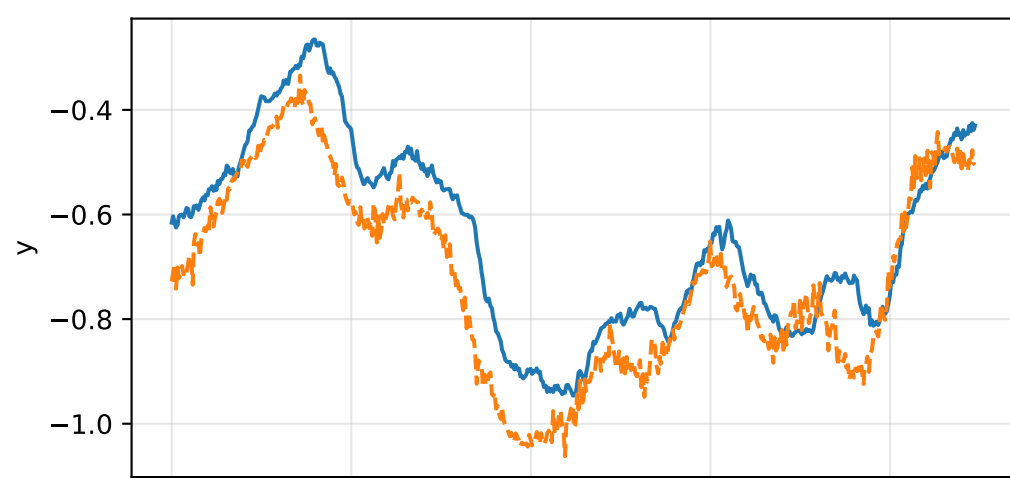
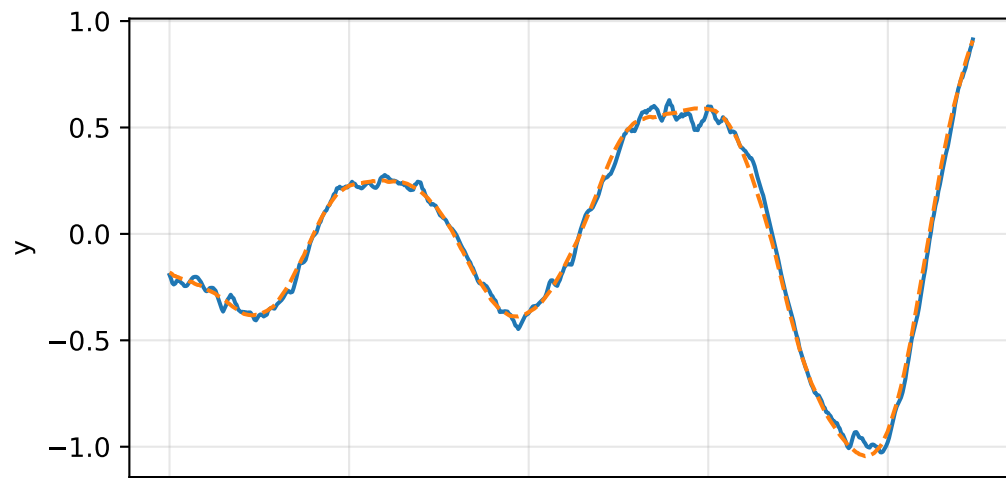
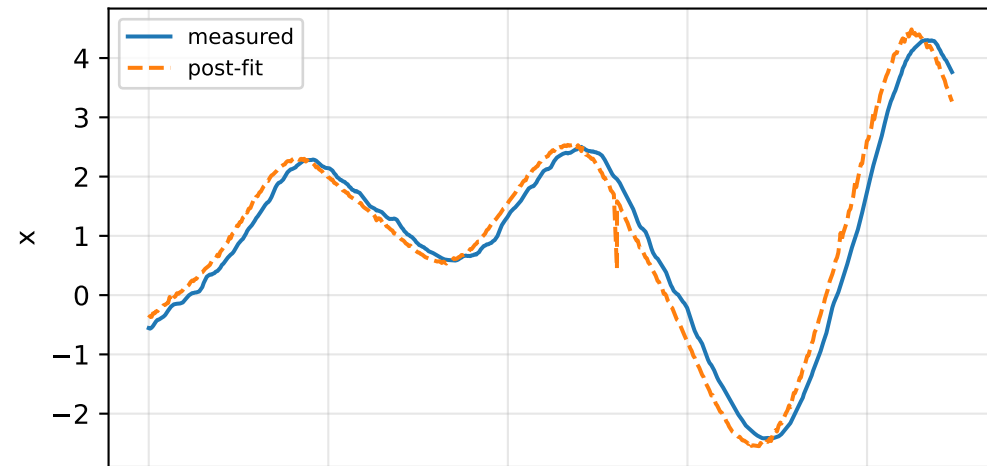


kkt_disabled: IMU comparison (diagnostic only)

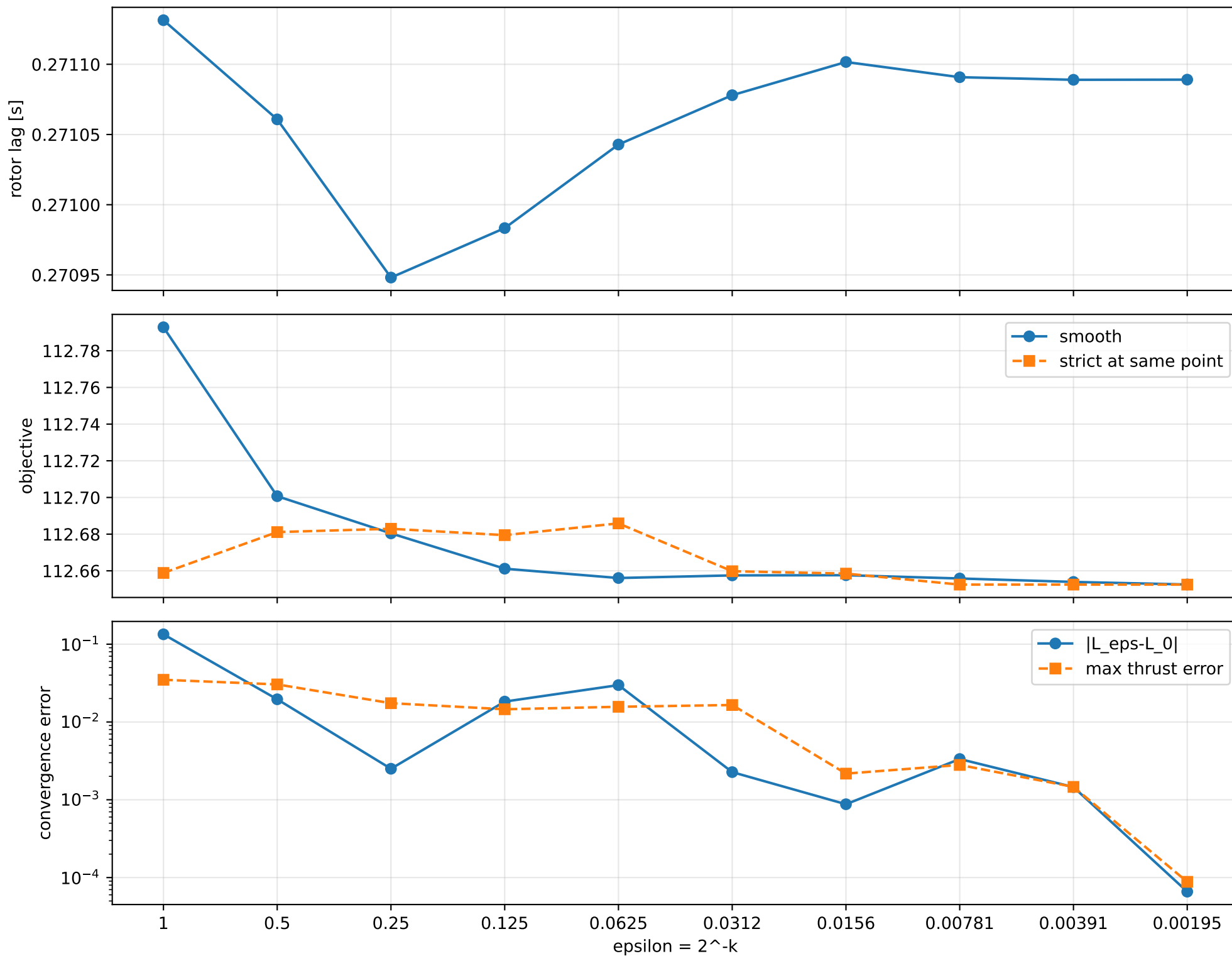
gyro [rad/s]



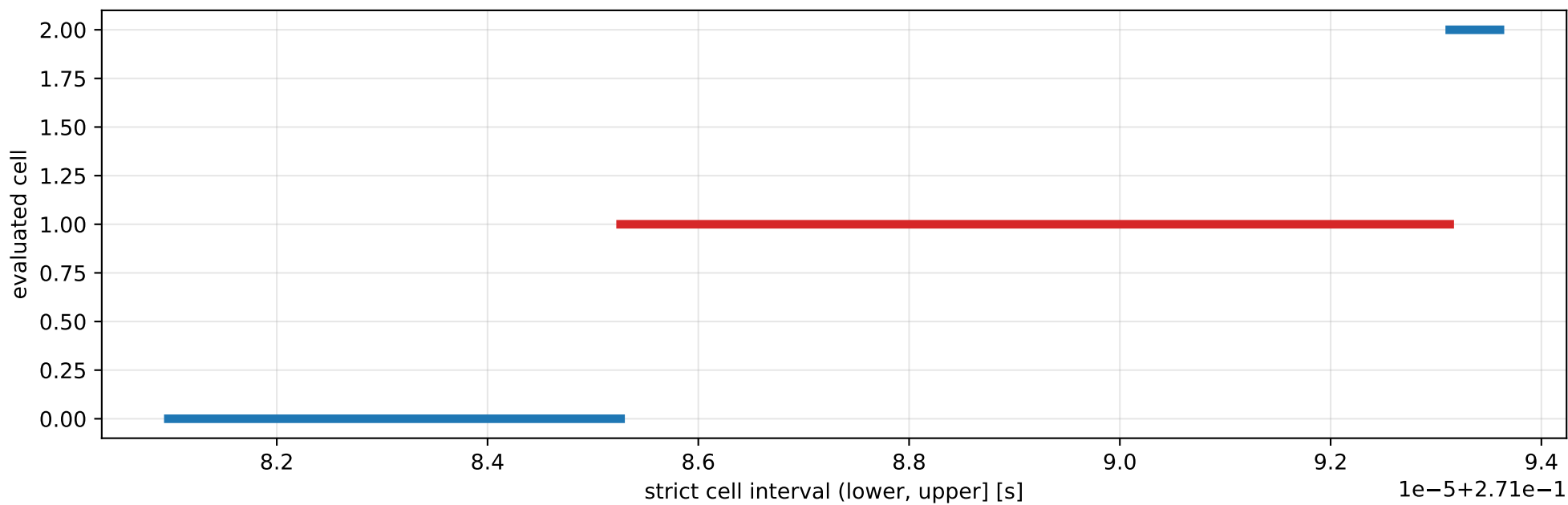
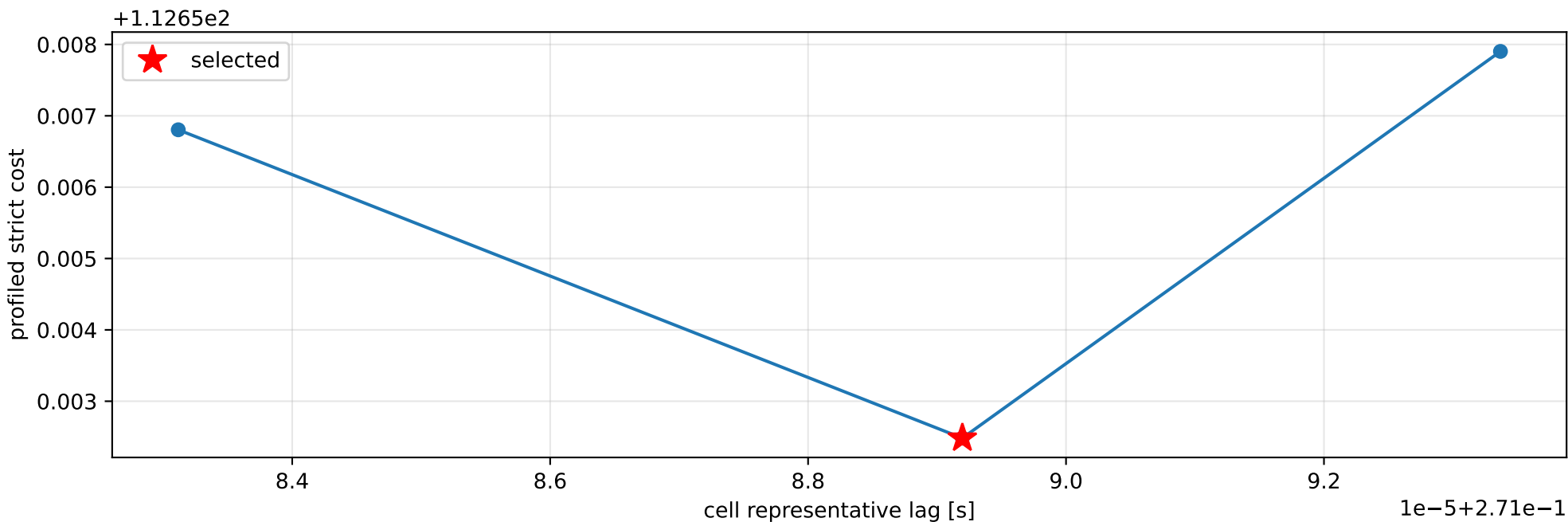
specific force [m/s^2]



kkt_disabled: smooth-to-strict continuation



kkt_disabled: exact strict-ZOH lag cells



selected (0.271085262, 0.27109313] s; final smooth 0.271089061 s; data boundary=False

kkt_disabled: parameter estimate

Case: kkt_disabled

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 7.52043445

inertia [kg m²]:

```
[[ 0.55443726  0.01372502 -0.00323582]
 [ 0.01372502  0.26168343 -0.08172247]
 [-0.00323582 -0.08172247  0.79125031]]
```

CoG body [m]: [-0.026435 0.01133929 0.00941535]

force effectiveness: [2.87131465 2.3238793 2.56341566 3.07927753]

scale-free J/m [m²]:

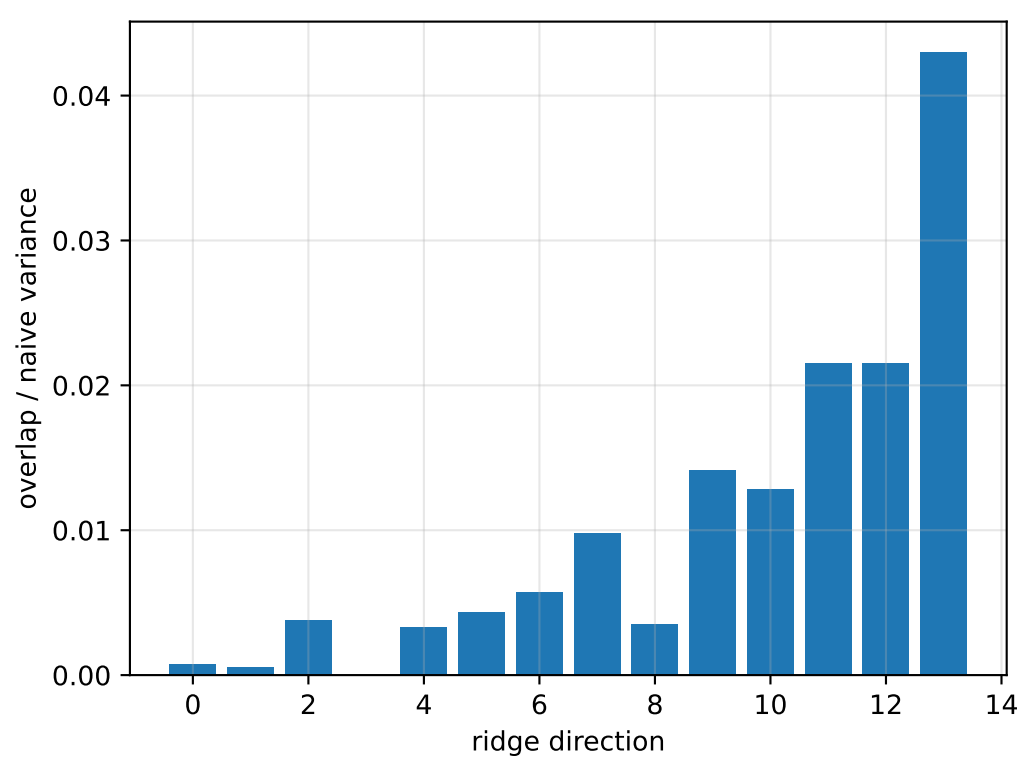
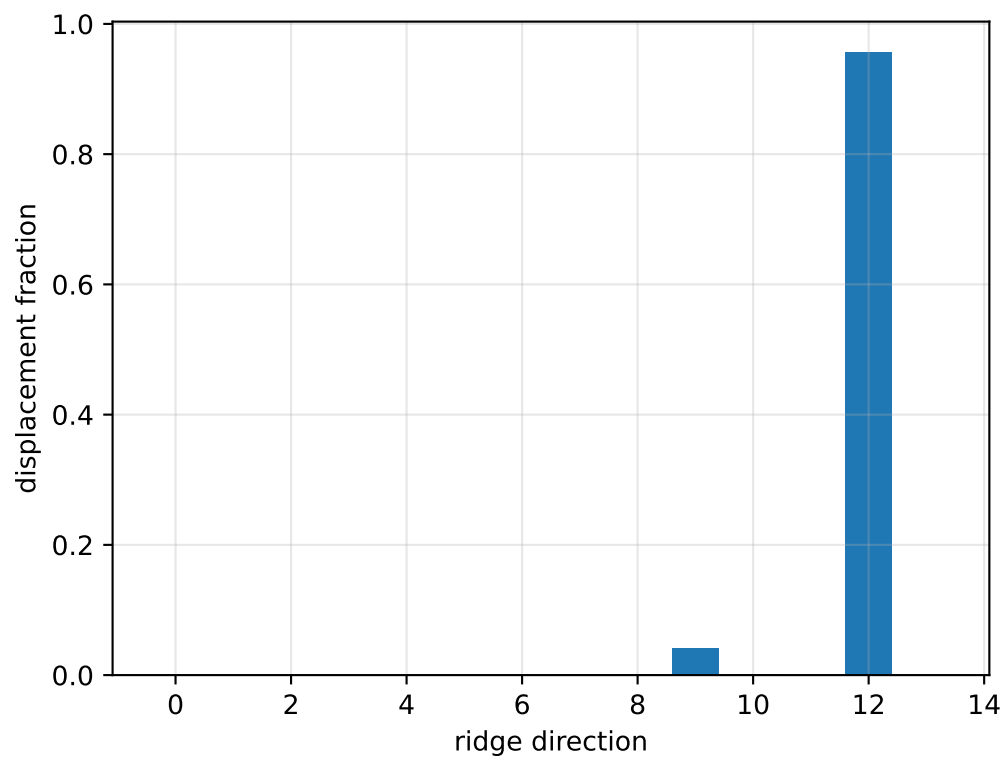
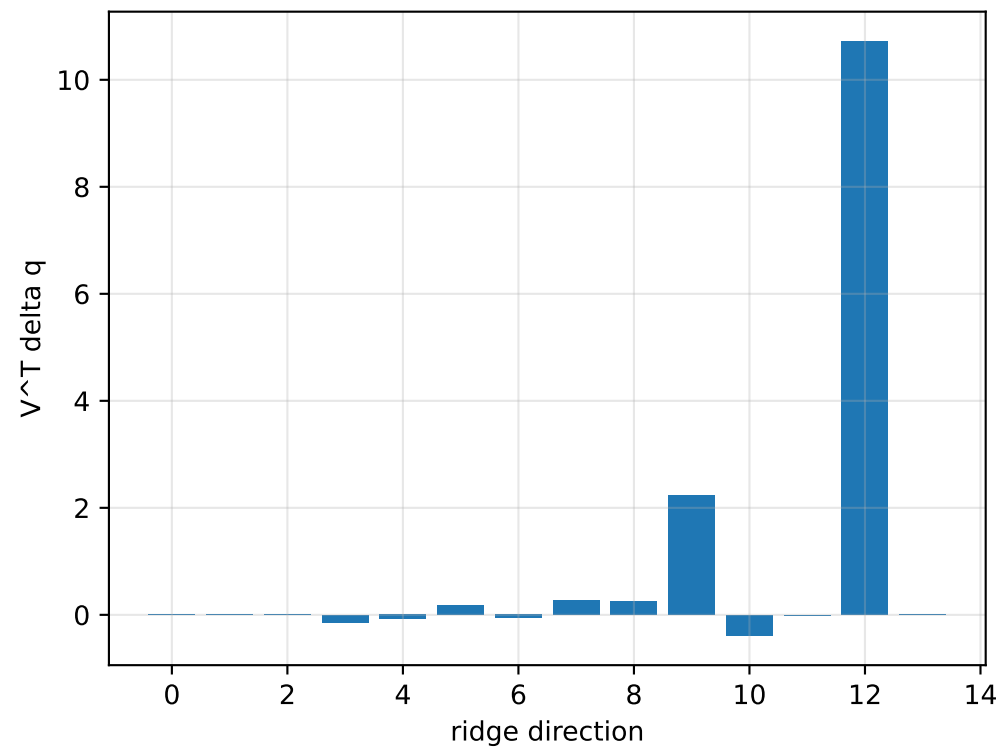
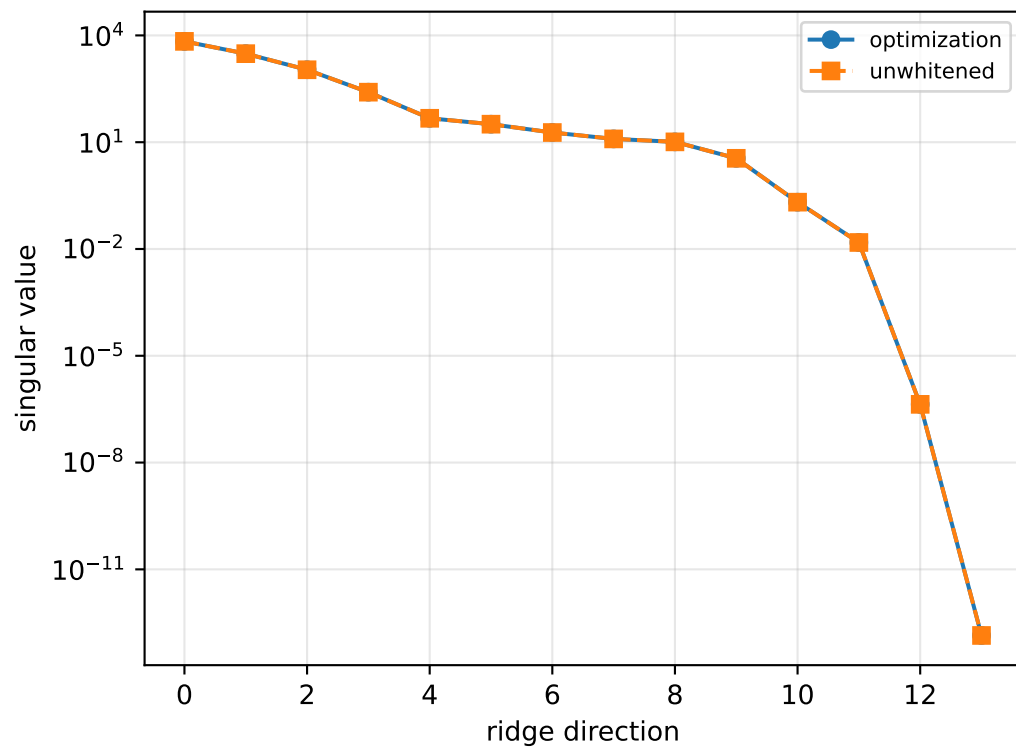
```
[[ 0.0737241  0.00182503 -0.00043027]
 [ 0.00182503  0.03479632 -0.01086672]
 [-0.00043027 -0.01086672  0.10521338]]
```

scale-free f/m: [0.3818017 0.30900865 0.34086005 0.40945474]

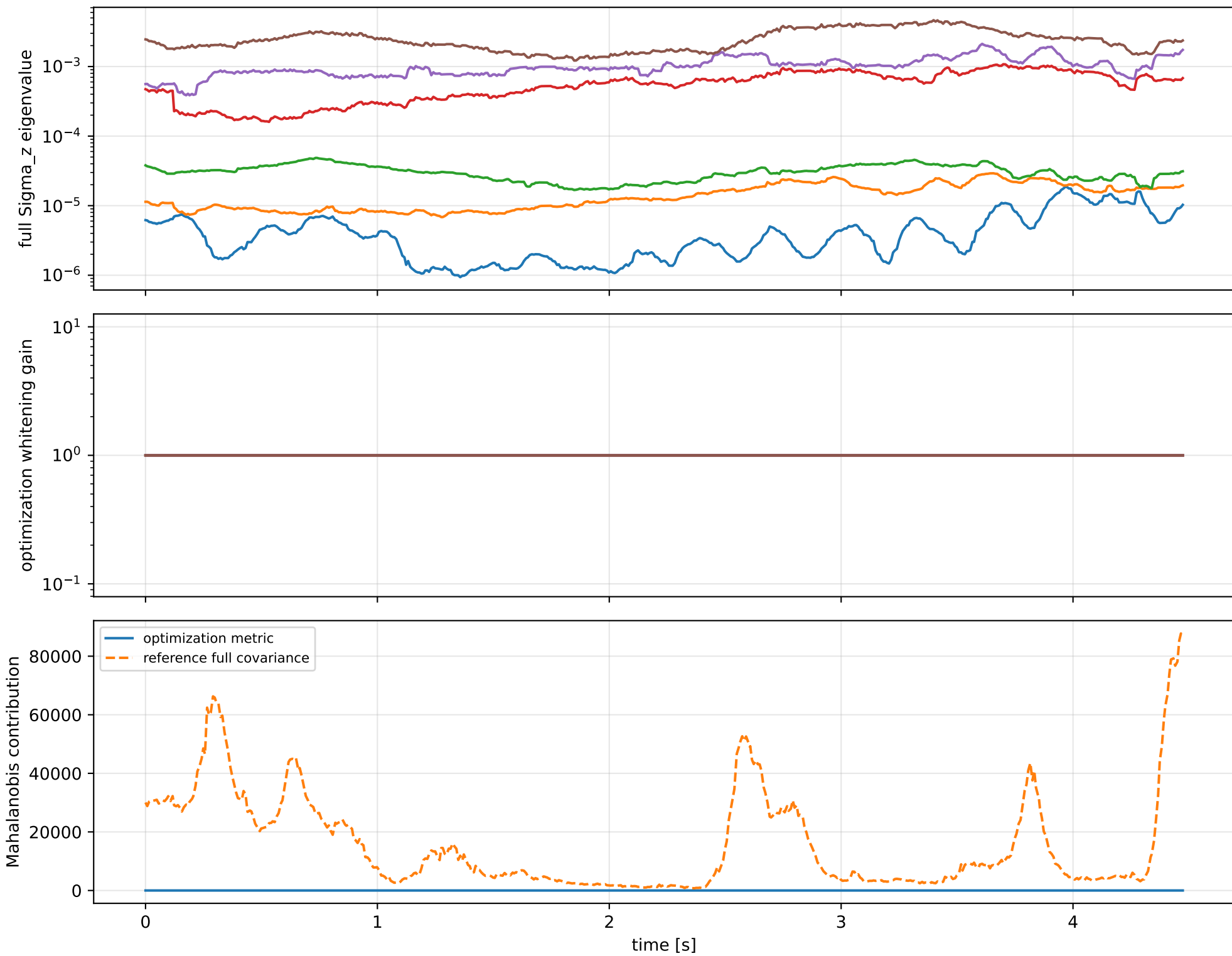
rotor lag cell: (0.271085262, 0.27109313] s

representative: 0.271089196 s

kkt_disabled: ridge spectrum (rank 13, nullity 1, || v ||=1.54e-13)



kkt_disabled: optimization vs reference covariance



kkt_disabled: wrench / closure (force max $2\text{e-}14$, torque max $4.3\text{e-}16$)

